ASCIA

SMART HOME SYSTEMS

TECHNICAL TENDERING

Request for Proposal — RFP

ASCIA OS System Development

Issuer	ASCIA Inc. — Montreal, Quebec, Canada
Document	RFP-2026-OS-001
Version	1.0 — Initial Document
Date of issue	February 2026
Response deadline	To be defined — 21 days after receipt
Project language	French (primary) — English accepted
Type of contract sought	Fixed price per phase + management fee for adjustments
Perimeter	ASCIA OS — Complete Home Automation Server System
Confidentiality	Confidential document — For the exclusive use of the recipient agency

— SECTION 01 —

Introducing ASCIA

Who we are, what we are building, what we are looking for

1. Introduction to ASCIA

1.1 Who is ASCIA

ASCIA is a Quebec-based company specializing in advanced home automation. We design, install, and maintain smart home systems for the residential, hospitality, and commercial markets in Canada. Our approach is entirely vertically integrated: we manufacture our own hardware, deploy our own software, and provide installation and ongoing support for each client.

The areas we cover include smart lighting, security and alarms, motorized blinds and curtains, locks and gates, water management, home theater and audio systems, video surveillance, and all connected IoT devices.

1.2 The Project — ASCIA OS

We are developing ASCIA OS, a complete home automation operating system designed to be installed on a dedicated server provided to our clients. ASCIA OS is based on Home Assistant OS (Apache 2.0 license — fork allowed) and features an immersive, real-time 3D interface, complete Plug & Play deployment, and local artificial intelligence.

This system is the core of our commercial offering. Each ASCIA customer receives a server with ASCIA OS pre-installed and simply needs to connect their devices to have a complete and operational home automation system.

> ASCIA OS is not a typical website or mobile application project. It's a complex embedded system combining a Linux OS, industrial IoT protocols, real-time 3D rendering, local artificial intelligence, and advanced network management. We are looking for an agency that understands this complexity and has the technical maturity to manage it.

1.3 What We Are Looking For

We are looking for a long-term technical partner—not a one-off service provider. The selected agency will be involved in the development of the MVP (Phase 1, 4-6 months), and then in the product evolution phases for a minimum of 18 months. We want an agency that:

- Understands complex systems and has already delivered technical products into real-world production
- Communicate transparently and proactively about blockages and risks
- Product of documented, tested, and maintainable quality code over the long term
- Treat our specifications with seriousness and ask the right technical questions
- Can mobilize a multidisciplinary team covering the 7 profiles described in this document

— SECTION 02 —

Technical Context

2. Technical Context of the Project

2.1 Overall Architecture

ASCIA OS is organized into three independent layers that communicate via well-defined APIs:

Layer	Technology	Role	MVP Responsibility
Layer 1 — Basic OS	Home Assistant OS (HAOS) Buildroot Linux	Embedded operating system, Docker supervisor, HA Core home automation engine	Yes — ASCIA OS image to be produced
Layer 2 — ASCIA Add-ons	Docker containers managed by HAOS Supervisor	12 independent ASCIA add-ons covering network, AI, NVR, audio, presence, etc.	Yes — MVP add-ons to be developed
Layer 3 — Frontend	SvelteKit + TypeScript Three.js / ebGL	Full ASCIA web interface replacing Lovelace (HA frontend	Yes — 3D interface + dashboard

2.2 Imposed Technical Stack

These technological choices are non-negotiable. They were defined after a thorough analysis of the project's constraints. The agency must master these technologies or demonstrate its ability to master them quickly.

Domain	Imposed Technology	Required Level	Reason for Choice
embedded OS	Home Assistant OS (HAOS)	Expert	Proven open-source database — 700k+ lines tested
Server add-ons	Python 3.11+ async + Docker	Expert	Native language of HA Core — essential
Web interface	SvelteKit + TypeScript	Expert	Maximum performance, native responsiveness
3D Engine	Three.js + WebGL	Expert	Industry standard, GPU-accelerate
Real-time communication	WebSocket (HA WS API protocol)	Expert	Standard HA, bidirectional, low latency
Zigbee protocol	Zigbee2MQTT	Confirmed	The most comprehensive, 3000+ devices
Local voice AI	Whisper (local OpenAI)	Confirmed	100% local, multilingual processing
Local LLM	Ollama + Mistral/Llama3	Confirmed	Cloud-free local inference
AI NVRs & Cameras	Frigate	Confirmed	Object detection, ONVIF, native RTSP
VPN	WireGuard	Confirmed	Modern, fast, automatable configuration
Database	PostgreSQL + SQLite	Expert	History, log, ASCIA data
Mobile application	Flutter + Dart	Expert	Cross-platform iOS/Android, single codebase
CI/CD	GitHub Actions + Docker Registry	Confirmed	Automated build and deployment

2.3 IoT Protocols to Integrate

Protocol	Technology	MVP Priority	Technical Difficulty
Zigbee	Zigbee2MQTT + USB coordinator	Critical	High — mesh network, 3000+ devices
PoE (ASCIA devices)	LLDP detection + Omada/Unifi switch API	Critical	High — Proprietary switch APIs
Wi-Fi IoT	mDNS + SSDP + UPnP scan	Critical	Average
ONVIF / RTSP (cameras)	ONVIF scan, RTSP stream extraction	Critical	Average
MQTT	Broker Mosquitto + ASCIA topics	Critical	Weak — well documented
KNX	xknx Python + import .knxproj	Critical	Very high — complex pro protocol
Z-Wave	Z-Wave JS UI	Average	Average
WHEAT	Permanent BLE scanner (presence)	Average	Average
UWB	Trilateration of anchors + tags	Average	Very high — Phase 2

2.4 ASCIA Add-ons to Develop

Add-on	Main Role	Key Technologies
ascia-frontend	SvelteKit Server — Full ASCIA Web Interface + 3D	SvelteKit, Three.js, WebSocket client
ascia-network-manager	Automatic VLAN creation, adaptive firewall, Omada/Unifi API	Python, iptables/nftables, REST switch API
ascia-device-manager	Self-discovery, onboarding, automatic naming	Python, mDNS, ONVIF, LLDP, MQTT
ascia-nvr	Native NVR + integrated Frigate + camera management	Python, Frigate, RTSP, HLS, FFmpeg
ascia-ai	Local AI: Whisper + Ollama + recognition	Python, Whisper, Ollama, InsightFace, CUDA
ascia-presence	UWB + BLE + mmWave — presence management	Python, BLE scan, trilateration algorithms
ascia-audio	Multiroom audio manager (Music Assistant)	Python, WebSocket, Spotify/Deezer APIs
ascia-vpn	WireGuard auto-configured + remote access management	WireGuard, Python, PKI key management
ascia-backup	Encrypted backup + restore + retention	Python, AES-256, rsync, PostgreSQL dump
ascia-diagnostics	Self-healing + proactive alerts + self-test	Python, system monitoring, alerts

— SECTION 03 —

Required Profiles

The 7 Essential Expertise Areas for ASCIA OS

3. Required Technical Profiles

These profiles represent the minimum required skills. The agency must submit actual CVs with concrete evidence (GitHub, projects, code) for each profile. Claims without evidence will not be considered.

3.1 Profile 1 — Senior Python Backend Architect

Criticality:MAXIMALE — This profile is the most important of the entire team.

Skill	Required Level	Expected Evidence
Python 3.10+ async (asyncio)	Expert — indispensable	Examples of async code in production
Docker microservices architecture	Expert	Projects with multiple containers in production
REST + WebSocket APIs (design)	Expert	Documented OpenAPI/Swagger APIs delivered
Linux system (processes, systemd, filesystem)	Confirmed	System administration or embedded Linux experience
PostgreSQL + SQLite (modeling, optimization)	Confirmed	Complex database schemas, migrations
MQTT (Mosquitto, topics, QoS, retained)	Confirmed	IoT project with MQTT in production
Security: TLS, JWT, AES encryption	Confirmed	Implementation in a real-world project
Home Assistant Core (internal architecture)	Appreciated but not a problem	4-week dedicated training accepted

3.2 Profile 2 — Senior Frontend Developer SvelteKit + Three.js

Criticality:MAXIMALE — The rarest profile. The 3D engine is the visual differentiator of ASCIA.

Skill	Required Level	Expected Evidence
SvelteKit + TypeScript	Expert	SvelteKit application in production (not just the basic Svelte version)
Three.js + WebGL (scene, models, animations)	Expert — indispensable	Functional interactive 3D demo to present
WebSocket client (real-time reactive states)	Expert	App with persistent WebSocket + automatic reconnection
Web performance (60 FPS, memory leaks, bundle)	Expert	Actual performance measurements in production
Advanced CSS + GPU-accelerated animations	Confirmed	Interface with smooth animations, 60 FPS
Loading 3D models (GLTF, GLB, OBJ)	Expert	Interactive 3D models loaded in real time
Real-time responsiveness (200+ simultaneous entities)	Confirmed	Application with many states updated simultaneously

3.3 Profile 3 — Senior Flutter Developer with Native Plugins

Criticality:HIGH — The mobile app is the primary touchpoint for ASCIA customers.

Skill	Required Level	Expected Evidence
Flutter + Dart (architecture, Riverpod, GoRouter)	Expert	Flutter apps in production on both stores
Native Flutter Plugins (Swift iOS + Kotlin Android)	Expert — indispensable	Custom plugin developed and published or in production
flutter_inappwebview (bridge JS↔Bidirectional Dart)	Confirmed	WebView hybrid app with functional JS injection
iOS (APNs) + Android (FCM) push notifications	Expert	Critical notifications, with actions, deep links
Geolocation background iOS/Android	Confirmed	Geofencing in production, iOS restrictions managed
ARKit (iOS) or ARCore (Android)	Appreciated	Spatial scanning or AR in a production app
Background processing (BLE, GPS, periodic tasks)	Confirmed	App with background services on both platforms

3.4 Profile 4 — Network & Security Engineer

Criticality:HIGH — Automatic network isolation is ASCIA's central security argument.

Skill	Required Level	Expected Evidence
Advanced TCP/IP networking (802.1Q VLAN, routing, NAT)	Expert	Configuring VLANs in production on real equipment
Linux Firewall (iptables / nftables)	Expert	Complex firewall rules in production
Omada TP-Link API + Unifi Ubiquiti API	Confirmed	Configuration script via these APIs (code to be submitted)
WireGuard (configuration, PKI, key rotation)	Expert	WireGuard VPN deployment in production
mDNS / Avahi / DHCP (IoT local network)	Confirmed	IoT network management with mDNS discovery
Application security (OWASP, basic penetration testing)	Confirmed	Security audit or report on a real project
Linux networking (bridges, virtual interfaces, namespaces)	Confirmed	Advanced Linux network configuration in production

3.5 Profile 5 — AI & Machine Learning Engineer

Criticality:HIGH — Joined in last Phase . Can be outsourced by a consultant if the agency does not have it internally.

Skill	Required Level	Expected Evidence
Python ML stack (PyTorch, ONNX, Hugging Face)	Expert	AI model deployed in production (not just a Jupyter notebook)
Whisper (optimization, model selection, latency)	Confirmed	Whisper local deployment with real-time latency measurements
Local LLMs via Ollama (Mistral, Llama3, Phi-3)	Confirmed	RAG or local chatbot deployed in production
GPU computing (CUDA, GGUF/GPTQ quantization)	Confirmed	GPU-optimized inference with performance metrics
Local Text-to-Speech (Piper TTS or Coqui TTS)	Appreciated	Local French text-to-speech deployed in an application
Classic machine learning for patterns (scikit-learn)	Confirmed	Pattern detection model deployed in production
AI deployment on limited hardware (without high-end GPUs)	Expert	Model optimization for mid-range CPUs or GPUs

3.6 Profile 6 — IoT & Protocols Specialist

Criticality:HIGH — Perhaps an external consultant if the agency does not have one internally, especially for KNX.

Skill	Required Level	Expected Evidence
Zigbee (mesh topology, coordinator, routers, LQI)	Expert	Zigbee network of 50+ devices managed in production
Zigbee2MQTT (advanced configuration, custom converters)	Expert	Zigbee2MQTT converter developed or contribution to the project
MQTT (topics, QoS, Mosquitto, bridges)	Expert	Multi-tier MQTT architecture in production
ONVIF + RTSP (IP cameras, stream extraction)	Confirmed	Integration of multi-brand cameras via ONVIF
KNX (TP bus, group addresses, DPT, xknx Python)	Confirmed	Project with real KNX integration (not just theoretical)
BLE and Z-Wave radio protocols	Confirmed	IoT project with BLE or Z-Wave in production
Embedded systems (datasheets, basic UART/I2C/SPI)	Appreciated	Interaction with physical IoT hardware

3.7 Profile 7 — Technical Project Manager

Criticality:HIGH — Our daily contact. Must be bilingual French/English preferably and sufficiently technical.

Skill	Required Level	Expected Evidence
Agile Scrum methodology (real, not theoretical)	Expert	Projects delivered using Agile methodology with sprint documentation
Sufficient technical understanding	Confirmed	Capable of reading and understanding Python and JavaScript code without coding
Impeccable written French communication	Expert	All reports and summaries in French
Proactive risk management	Expert	Examples of risks identified and addressed before they became crises
Tools: Jira/Linear, GitHub, Slack/Teams	Confirmed	Daily use of these tools on past projects
Availability overlapping hours Montreal	Expert	Common time slot: 9am-1pm Montreal time minimum

Selection Criteria

How We Evaluate Your Agency

4. Agency Selection Criteria

4.1 Evaluation Grid

Each bidding agency will be evaluated on the following criteria, weighted according to their importance to the success of the ASCIA OS project.

Evaluation Criteria	Weight	What We Evaluate Specifically
Technical quality of the proposal	25%	Understanding the architecture, identifying risks, and making justified choices.
Concrete evidence of skills	25%	GitHub code, working demos, similar projects in production
Quality and realism of the estimate	20%	Hourly breakdown, justifications, honest estimates
Available and verifiable profiles	15%	Detailed CVs with GitHub + portfolio for each required profile
Work process and communication	10%	Frequency of deliverables, tools, availability
Verifiable customer references	5%	Contactable clients who attest to the quality of the deliverables

4.2 Immediate Elimination Criteria

If one of these criteria is not met, the application is automatically eliminated, regardless of the quality of the other elements.

Elimination Criterion	For what
No presentable public or private code for key profiles	Without proof of actual code, it is impossible to assess the true technical quality.
Estimate delivered in less than 72 hours after receipt of specifications	A proper quote for a project of this complexity requires a minimum of 5 days of analysis.
The proposal makes no mention of technical risks.	A partner who doesn't see the difficulties cannot manage them.
No profile capable of creating native Flutter (Swift/Kotlin) plugins	This skill is essential for the native functionalities of the mobile app.
No developer capable of working on embedded Linux and Docker	The ASCIA OS image and add-ons absolutely require this expertise.
Request for more than 25% of the total budget as an initial deposit	Reputable partners charge upon delivery — not upon signature
Refusal to begin with a mini validation contract (Phase 0)	If the agency is not confident in proving its value on a test module, that's a signal.
No availability for minimal time overlap with Montreal	Coordination requires a minimum of 3 hours of simultaneous work per day

4.3 What We Particularly Value

- A proposal that identifies risks we haven't mentioned — that's a sign of a team that has actually read and analyzed the specs
- Relevant technical questions before submitting the proposal
- A portfolio with readable, commented, and well-structured code on GitHub
- An experience with real-time systems (IoT, finance, gaming, medical) — not just websites
- A culture of documentation — delivered projects must have real technical documentation
- Customer testimonials that talk about code quality and communication, not just deadlines

— SECTION 05 —

Technical Questions

The Comprehension Test You Must Take

5. Technical Questions — Required Answers

> These questions must be answered in detail in your proposal. Vague or generic answers will be considered an inability to answer. There is no such thing as a bad, honest answer—there are only honest answers and answers that attempt to conceal a lack of competence.

5.1 Questions about System Architecture

Question 1 — Managing Home Assistant Updates

ASCIA OS is based on Home Assistant OS. Home Assistant releases major updates approximately every four weeks, and security updates more frequently. Some of these updates may modify internal APIs that your add-ons rely on.

Describe precisely your strategy for: (a) freezing the HA version for production stability, (b) testing new HA versions without breaking existing client installations, (c) migrating ASCIA add-ons when an HA update is incompatible.

Expected answer: *Versioning strategy, regression testing pipeline, migration policy. Minimum 200 words.*

Question 2 — Physical Hardware Tests for IoT

ASCIA OS must work with real physical devices: Omada and Unifi PoE switches, USB Zigbee coordinators, ONVIF IP cameras, mmWave sensors, KNX modules. These devices cannot be fully simulated in a virtual environment.

How will you: (a) procure or access the necessary test equipment, (b) structure your hardware integration tests, (c) manage differences in behavior between brands of devices using the same protocol?

Expected answer: *Concrete plan for accessing equipment, hardware testing strategy, incompatibility management. Minimum 150 words.*

Question 3 — 3D Engine Performance

The ASCIA 3D plan must display the real-time state of 200+ entities simultaneously (RGB lights, doors, blinds, human presence) at a stable 60 FPS in a Chrome browser, on a computer without a dedicated GPU (Intel or AMD integrated graphics processor).

Describe your approach to: (a) optimize Three.js rendering for this performance constraint, (b) handle the updating of 200+ states simultaneously via WebSocket without blocking the main thread, (c) implement an automatic degraded mode if performance is insufficient.

Expected answer: *Specific techniques: instancing, Level of Detail, Web Workers, requestAnimationFrame, dirty checking. Minimum 200 words.*

Question 4 — Network Self-Discovery Architecture

ASCIA OS must automatically detect any device connected to the network — whether it is PoE, Zigbee, Wi-Fi, or ONVIF — and integrate it into the system in less than 30 seconds, without manual intervention from the client.

Explain: (a) how you will implement automatic PoE detection via the Omada/Unifi API, (b) how you will handle conflicts when the same device is detectable by multiple mechanisms (e.g., a PoE + ONVIF + mDNS camera), (c) how you will automatically identify the ASCIA device type from its network profile.

Expected answer: *Architecture of the discovery service, duplicate management, profile system. Minimum 200 words.*

Question 5 — Automatic Network Isolation

When a customer connects a new device to the PoE switch, ASCIA OS must automatically create or update VLAN and firewall rules to isolate that device, without touching the customer's existing modem or router.

Describe: (a) how you will interact with the Omada API and the Unifi API to programmatically create VLANs and firewall rules, (b) what your strategy is if the client switch is neither Omada nor Unifi, (c) how you will handle rollbacks if a firewall rule accidentally cuts off connectivity to a critical device.

Expected answer: *Concrete implementation of the switch APIs, fallback strategy, secure rollback mechanism. Minimum 200 words.*

5.2 Questions about Processes and Quality

Question 6 — Managing the HA Learning Curve

We understand that your team may not have prior experience with Home Assistant. We accept this reality provided it is addressed proactively.

Describe: (a) your concrete onboarding plan for your developers to master the HA Core architecture in 4 weeks, (b) who in your team will be responsible for this skills development and how you will measure it, (c) your strategy for accessing external HA expertise if you get stuck on an architecture problem.

Expected answer: *Detailed week-by-week onboarding plan, progress metrics, identified HA expert contacts. Minimum 150 words.*

Question 7 — Code Quality and Documentation

ASCIA OS will need to be maintained and evolved for 5 to 10 years. The quality of the initial code determines the ease of future maintenance.

Explain: (a) your Python code standards (linting, typing, test coverage), (b) how you document code during development and not after, (c) how you manage code reviews and who has the final say on quality, (d) a concrete example of a quality issue you detected and fixed in a past project.

Expected answer: *Concrete standards with tools (Black, mypy, pytest, coverage %), review process, real-world example. Minimum 200 words.*

Question 8 — Project Risk Management

Identify the 5 most significant technical risks you see in this project, and for each one, describe the mitigation strategy you propose.

We want to see your level of understanding of the project. A partner who identifies the same risks as us—or risks we hadn't identified—demonstrates that they have truly analyzed the specifications.

Expected answer: *A minimum of 5 risks, including: risk description, estimated probability, impact, and concrete mitigation strategy. Minimum 300 words.*

Question 9 — Communication and Managing Jet Lag

Our team is based in Montreal (UTC-5). Your agency is likely in a different time zone. Daily coordination is critical on a project of this complexity.

Please provide: (a) a concrete communication plan (frequency of standups, tools, times), (b) your protocol for urgent blockages that require a quick response even outside common hours, (c) how you will involve us in architectural decisions without creating dependencies that slow down your team.

Expected answer: *Detailed weekly schedule, specific tools, emergency protocol. Minimum 150 words.*

Question 10 — Crisis Scenario

Scenario: We're in month 3 of the project. Your senior frontend developer—the one who's an expert in Three.js and who built the entire 3D engine—announces their resignation with two weeks' notice. The 3D rendering is 60% complete but not yet functional.

Describe specifically: (a) what you do in the first 48 hours, (b) how you ensure that the knowledge of the departing developer is transferred, (c) what profile you mobilize as a replacement and how long it will take, (d) the impact on the schedule and budget, and how you communicate this to us.

Expected answer: *Concrete action plan with deadlines, knowledge transfer procedure, available replacement profile, quantified impact. Minimum 200 words.*

— SECTION 06 —

Expected Deliverables

What Your Proposal Should Contain

6. Content of the Expected Proposal

An incomplete proposal will be automatically disqualified. Each point below is mandatory. We prefer an honest proposal that acknowledges shortcomings to one that claims to have everything figured out.

6.1 Mandatory Documents to Provide

Document	Required Content	Format
1. Presentation of the agency	History, size, current clients, technical specialties, certifications	PDF — 5 to 10 pages
2. Answers to the 10 technical questions	Detailed answers to each of the questions in Section 5	PDF — numbered sections
3. CVs of the proposed profiles	A CV per profile is required, including a personal GitHub link and a list of projects.	Individual PDF per profile
4. Portfolio of similar projects	A minimum of 3 projects, including: technical description, agency role, measurable results, client contact, and reference.	PDF with screenshots and architecture
5. Detailed MVP Estimate	Module-by-module breakdown with hours, assigned profile, hourly rate	Excel or detailed spreadsheet
6. Proposed schedule	Sprint schedule spanning 4-6 months with deliverables at each stage	Gantt chart or sprint table
7. Contract proposal	Proposed payment structure, PI clauses, exit clause, SLA	PDF or Word
8. Technical demo (optional but highly appreciated)	Any relevant technology demo: Three.js 3D, native Flutter plugin, IoT system, Docker add-on	Video link or live demo

6.2 Minimum Technical Evidence

Beyond the documents, we require the following concrete evidence for critical profiles:

Profile	Minimum Evidence Required	How to Present It
Frontend Three.js	A working interactive 3D demo in a browser	Public URL or screencast video
Async Python backend	Example code for an async Python service with tests	GitHub repository (public or private access provided)
native flutter	A custom Flutter plugin with native Swift or Kotlin code	GitHub repository or test APK/IPA
Network Engineer	VLAN or firewall configuration script via API (Omada/Unifi or similar)	Commented Python or bash code
IoT Engineer	Integration of an IoT protocol (Zigbee, MQTT, or other) into a real system	GitHub repository + project description

6.3 Proposed Payment Structure

We propose the following payment structure as a basis for negotiation. The agency may suggest adjustments but must justify them.

Milestones	% of the Budget	Trigger
Initial deposit (onboarding + setup)	15%	Contract signing and start of MVP
Full MVP delivery	25%	All MVP modules delivered, tested and documented
MVP stability	20%	30 days without a critical bug on a real client installation with MVP
Final product (V1)	25%	full ascia os v1 delivered, tested and documented
warranty and document	15%	90 days of stability and delivery with complete system documentation

— SECTION 07 —

Contract Terms

Intellectual Property, Confidentiality & Essential Clauses

7. Non-Negotiable Conditions and Clauses

The following clauses are non-negotiable. They protect ASCIA Inc. and ensure that the work produced remains our exclusive property. Any agency that cannot accept these conditions must not submit a proposal.

7.1 Intellectual Property

Clause	Detail
Property from creation	All code, design, architecture, and documentation produced within this project belong to ASCIA Inc. from its inception—not upon final delivery, not after full payment. From the very first commit.
No reuse	The agency may not reuse, adapt or draw inspiration from the ASCIA code for other clients, even after the end of the contract.
ASCIA Repository	All code is committed to GitHub repositories owned by ASCIA Inc. The agency does not have access to the repositories after the end of the contract.
Included deliveries	Deliverables include: commented source code, technical documentation, architecture diagrams, configuration files, and deployment scripts. Nothing is delivered in binary format only.

7.2 Confidentiality

Clause	Detail
NDA prior to sharing specs	A non-disclosure agreement (NDA) must be signed before the detailed specifications are shared.
No unapproved subcontractors	The agency cannot subcontract parts of the project to third parties without prior written agreement from ASCIA Inc.
Post-contract confidentiality	The confidentiality obligations apply for 5 years after the end of the contract.
No public reference without agreement	The agency may not mention ASCIA as a client in its portfolio or marketing communications without written agreement from ASCIA Inc.

7.3 Quality and Deliverables

Clause	Detail
Continuous access to the code	ASCIA Inc. has access to GitHub repositories at all times, including branches under development. No code is hidden until delivery.
Quality standards	Minimum 70% unit test coverage for backend code. Mandatory documentation for each ASCIA add-on. Mandatory code review before merge.
Profile replacement	If a key member of the team leaves, the agency must replace them with a member of equivalent skill level within a maximum of 15 business days. The cost of knowledge transfer is borne by the agency.
Response latency	Any critical issue or problem must receive an initial response within 4 hours during business hours. Non-urgent questions must be answered within 24 hours.

7.4 Exit Clause

ASCIA Inc. reserves the right to terminate the contract with 30 calendar days' notice, at any time and for any reason. Upon termination, the agency must deliver all product code produced up to that point, fully documented, within 10 business days. Payment for deliverables completed and validated up to the termination date is due from ASCIA Inc. within 30 days.

— SECTION 08 —

Response Instructions

Process, Deadlines & Contact

8. Instructions for Responding to this Invitation to Tender

8.1 Selection Process

Stage	Action	Deadline
1 — Reception and NDA	Confirm receipt of this RFP and sign the NDA to receive the complete specifications.	Immediately upon reception
2 — Clarification questions	Send your technical questions about the project before preparing your proposal.	Within 7 days of receipt
3 — Submission of the proposal	Submit the complete proposal including all documents from Section 6	13 days after receipt of the RFP
4 — Call for applications	60-minute presentation of your proposal with the proposed technical team	Within 14 days of receipt
5 — Final Decision	Selection of the partner agency based on the overall proposal	20 days after the receipt

8.2 What You Need to Do Now

1. Confirm by email the receipt of this document and your intention to submit a proposal.
2. Sign the NDA to receive the complete specifications (ASCIA OS specifications and ASCIA Mobile specifications).
3. Identify the 7 required profiles within your team and prepare their CVs with supporting evidence.
4. Prepare your technical questions — we prefer pointed questions to a proposal that makes assumptions.
5. Start the technical analysis of the specs before preparing your estimate

8.3 What Will Be Provided After Signing the NDA

- CDC ASCIA OS Complete — 24 sections, 78 pages — Detailed technical specifications of the server system
- CDC ASCIA Mobile Complete — Flutter iOS/Android Application Specifications
- List of 179 features with MVP/Post-MVP/V2.0 classification and effort estimates
- List of physical devices you'll need (switches, Zigbee coordinators, cameras)

8.4 Criteria for a Good Proposal

> We won't necessarily select the cheapest agency. We'll select the agency that best demonstrates an understanding of the project, the ability to deliver quality work, and a commitment to transparency over the long term. An honest proposal that acknowledges shortcomings and offers a plan to address them is worth more than a proposal that's perfect on paper.

8.5 Frequently Asked Questions

Question	Answer
Do you need experience in home automation or Home Assistant?	No, but we need to be honest enough to say it and have a concrete plan to fill this gap.
Can we propose a different technology stack?	Yes if valid arguments and a thorough presentation
Can modules be outsourced to partners?	Only with prior written agreement. Each subcontractor must be presented and approved.
What is the minimum acceptable team size?	A minimum of 5 people are required to cover profiles 1, 2, 3, 4 and 7. Profiles 5 and 6 can be external consultants.
Is there a maximum budget?	We don't publish a maximum budget to avoid proposals based on that maximum. Be honest about your rates.
Can the project start before the end of the selection process?	Nothing starts before the finalists are selected.

8.6 Contact

Business	ASCIA Inc.
Address	Montreal, Quebec, Canada
Email subject	[RFP-2026-OS-001] + your agency name
Language	French preferred — English accepted

Thank you for your interest in this project.

ASCIA OS is an ambitious project seeking a partner who can match its ambition.